

Leakage-Aware Walk-Forward Machine Learning for Bitcoin

A Quantitative Research and Backtesting Study

Alexandre Sigiridis-Schlepp

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This report audits and evaluates a leakage-aware machine-learning framework for forecasting three classes of next-period Bitcoin returns—down, neutral, and up—from Bitcoin and cross-asset daily market data. The experiment uses 3,145 observations and 101 engineered features, a chronological 60/20/20 train/validation/test split, validation-only trading-threshold calibration, and a strict expanding-window walk-forward test with model retraining every 30 observations. The final ensemble combines regularized multinomial logistic regression and histogram gradient boosting. The test balanced accuracy is 0.3772, only modestly above the three-class reference level of one third, and the log loss is 1.0955 versus 1.0986 for uniform class probabilities. The long/cash strategy loses 3.75% over the test sample, compared with -19.24% for buy-and-hold and -7.97% for a simple trend filter, but is invested on only 1.59% of test days. Consequently, the lower cumulative loss is primarily evidence of low market exposure rather than established alpha. A more interesting result is that the model contains more information about movement regime than about direction: one-vs-rest AUC is 0.601 for the neutral class, while up and down AUCs are 0.527 and 0.523, respectively; conditional directional AUC on non-neutral observations is 0.484. The experiment therefore does not establish a robust directional edge, but it motivates a hierarchical regime–direction–risk architecture. A code audit also identifies an important residual issue: the initial training fit used to calibrate validation thresholds is not purged by two rows at the train/validation boundary, even though labels depend on two future opens. This is a small but real boundary leakage that should be corrected in the next revision.

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1. Introduction

1.1. Motivation

Financial machine learning is unusually easy to overstate. A model can appear successful because information from the future leaks into the feature pipeline, because strategy parameters are tuned on the final holdout period, because transaction costs are ignored, or simply because a low-exposure strategy avoids a falling market. Backtest selection itself can become a multiple-testing problem: repeated experimentation can turn historical data into an implicit optimization target (White, 2000; Bailey et al., 2017).

The project studied here was intentionally designed around those failure modes. Its purpose is not to claim that “AI predicts Bitcoin,” but to ask a narrower and testable question:

Can a leakage-aware, walk-forward machine-learning framework extract economically useful information from Bitcoin market data, and if so, is that information primarily directional or related to market regime and movement amplitude?

1.2. Scope and contributions

The supplied implementation downloads daily BTC and cross-asset market data, constructs only backward-looking features, defines a three-class next-tradable-return target, trains a regularized linear/nonlinear ensemble, calibrates trading thresholds on a validation set, then freezes those thresholds for a 629-observation walk-forward test. Costs and slippage are applied to changes in position. The present report contributes four things beyond merely repeating the program output:

1. a code-level audit of temporal causality, preprocessing, target maturity, execution timing and threshold selection;
2. an exact quantitative reconstruction of the reported out-of-sample performance;
3. additional diagnostics computed from the supplied prediction file, including class AUCs, probability-confidence statistics, a confusion matrix and a forensic analysis of the maximum drawdown;
4. an interpretation that separates directional skill from regime/movement information and separates nominal benchmark outperformance from genuine evidence of alpha.

1.3. Experimental pipeline

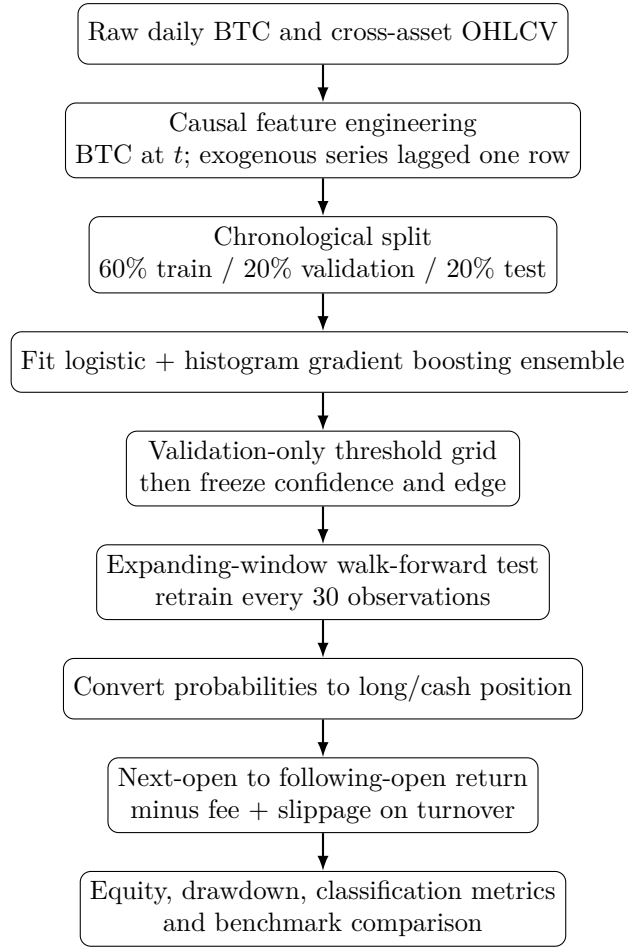


Figure 1.1.: Reconstructed experimental pipeline. The high-level structure is sound for time-series research because model fitting, strategy calibration and final evaluation are separated temporally. The audit in chapter 6 nevertheless identifies a two-row label-maturity leak at the initial train/validation boundary.

2. Financial Time-Series Prediction Challenges

2.1. Non-stationarity and low signal-to-noise ratio

Market return distributions and predictor relationships vary across volatility, liquidity, monetary and risk-appetite regimes. A classifier fitted to a long historical window therefore estimates an average mapping that may be locally wrong. The problem is compounded by the low signal-to-noise ratio of one-day financial returns: even a genuine weak signal can produce classification metrics close to naive baselines.

2.2. Leakage and validation design

Randomly shuffling time-series observations can make the training set contain observations that occur after validation or test observations. More subtly, a row whose *features* are dated before a split may still have a *label* that requires future prices after the split. This is exactly why label maturity and purging matter. Time-series evaluation must mimic the information set available at the historical decision time. Literature on time-series validation emphasizes that dependence and temporal structure require special care (Bergmeir et al., 2018).

2.3. Backtest overfitting

A final holdout loses its meaning if it is inspected repeatedly and design choices are then revised until performance improves. White's data-snooping framework and later work on backtest overfitting formalize this selection risk (White, 2000; Bailey et al., 2017). A single apparently good equity curve is therefore not a statistical proof of economic predictability.

2.4. Why accuracy alone is insufficient

A classifier can improve class accuracy without producing an implementable trading edge. Economic performance depends on which errors occur, on the return magnitude associated with those errors, on market exposure, turnover, costs and execution delay. Conversely, a low-exposure strategy can have a shallower drawdown than buy-and-hold even with no meaningful forecasting skill. The experiment therefore reports both classification and portfolio metrics.

3. Data

3.1. Assets

The configured primary instrument is **BTC-USD**, beginning from 1 January 2018 with no fixed end date. The seven exogenous series are listed in table 3.1.

Table 3.1.: Configured market inputs. Economic interpretations are motivations for inclusion, not evidence that the corresponding feature is predictive.

Ticker	Market	Motivation
BTC-USD	Bitcoin	Primary price, volatility, trend, volume and target series
ETH-USD	Ether	Crypto cross-market state and relative momentum
SPY	S&P 500 ETF	Broad equity risk appetite
QQQ	Nasdaq-100 ETF	Growth/technology risk appetite
GLD	Gold ETF	Defensive / macro cross-asset state
$\hat{VIX}$	CBOE volatility index	Equity-market stress / implied-volatility proxy
DX-Y.NYB	U.S. Dollar Index	Dollar regime
$\hat{TNX}$	U.S. 10-year yield index	Interest-rate environment

3.2. Observed dataset size and split

After feature construction and the code’s minimum-coverage filter, the saved run contains 3,145 rows and 101 features. The chronological split consists of 1,887 training rows, 629 validation rows and 629 test rows. The test prediction file spans 22 December 2024 through 11 September 2026. Exact persisted calendar boundaries for the training and validation sets are not stored in the supplied outputs, so they are not reconstructed by assumption.

3.3. Temporal alignment of exogenous markets

BTC trades continuously while SPY, QQQ, GLD, VIX, the dollar index and the 10-year yield follow traditional-market calendars. The implementation first reindexes each exogenous close onto the BTC calendar, forward-filling up to seven days, computes lagged/rolling statistics, and finally shifts every exogenous feature by one row. This deliberate one-day lag sacrifices freshness to reduce accidental same-session and time-zone look-ahead. It is conservative, but it does not fully establish timestamp equivalence across data vendors and market closes; this remains a synchronization limitation.

4. Feature Engineering

4.1. Causality

The BTC feature function uses rolling windows, exponentially weighted statistics and lags based on the current or earlier completed BTC bars. Its own docstring states that these features are known at the close of the current BTC daily bar. Exogenous features are shifted by one calendar row after alignment. Missing values are not globally filled before splitting; instead, median imputation occurs inside each fitted model pipeline.

4.2. Representative feature families

Table 4.1.: Representative feature families. The full 101-feature list is reported in Appendix B.

Family	Examples	Role
Returns / momentum	BTC log returns over 1,2,3,7,14,30,60,90 d; exogenous log returns	Directional persistence / reversal
Volatility	BTC rolling vol 7–60 d, ATR/price, range; exogenous 20 d vol	Movement amplitude and regime
Trend	Distances to EMA 10,20,50,100,200; MACD and histogram	Trend state
Oscillators	RSI(14), Bollinger and width	Relative position within recent range
Drawdown	BTC drawdown from 20,60,180 d rolling maxima	Stress / recovery regime
Volume	30 d z-score and 1/7 d log-volume changes	Participation / activity
Cross-asset	ETH, SPY, QQQ, GLD, VIX, DXY, TNX levels/returns/z-scores/volatility	Macro and risk regime
Relative	BTC minus ETH/SPY 30 d momentum	Relative strength
Calendar	Sine/cosine day-of-week and month	Low-dimensional seasonality

4.3. Mathematical definitions

The implementation uses log returns for many momentum features,

$$r_{t,h}^{\log} = \ln P_t - \ln P_{t-h} = \ln\left(\frac{P_t}{P_{t-h}}\right), \quad (4.1)$$

and simple close-to-close returns for rolling volatility,

$$r_t = \frac{P_t}{P_{t-1}} - 1, \quad \sigma_{t,N} = \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} (r_{t-i} - \bar{r})^2}, \quad (4.2)$$

where the code uses population standard deviation (`ddof=0`) for these features.

Distance from an exponential moving average is normalized by price scale,

$$d_{t,N}^{\text{EMA}} = \frac{P_t}{\text{EMA}_{t,N}} - 1. \quad (4.3)$$

The rolling drawdown features use finite lookback maxima rather than the all-time maximum,

$$DD_{t,N} = \frac{P_t}{\max_{t-N+1 \leq \tau \leq t} P_\tau} - 1, \quad (4.4)$$

for $N \in \{20, 60, 180\}$ days. ATR is computed from the exponentially smoothed true range and divided by close, while RSI(14) is centered and scaled to approximately $[-1, 1]$ through $(\text{RSI} - 50)/50$.

5. Prediction Target and Machine-Learning Models

5.1. Three-class target

For a feature row dated t , the target return is not the contemporaneous close-to-close move. It is the next tradable open-to-open return,

$$r_t^{\text{exec}} = \frac{O_{t+2}}{O_{t+1}} - 1. \quad (5.1)$$

The decision uses information through the BTC close at t ; execution begins at O_{t+1} and the position is held to O_{t+2} . This explicit delay is an important anti-look-ahead safeguard.

The neutral half-band is volatility dependent,

$$b_t = \max \left(0.004, 0.25 \sigma_{t,30}^{\text{past}} \right), \quad (5.2)$$

where $\sigma_{t,30}^{\text{past}}$ is the 30-day rolling standard deviation of past BTC close returns. The class is

$$y_t = \begin{cases} +1, & r_t^{\text{exec}} > b_t, \\ -1, & r_t^{\text{exec}} < -b_t, \\ 0, & \text{otherwise.} \end{cases} \quad (5.3)$$

The minimum neutral return is therefore 0.4%, with a larger band when one quarter of recent daily volatility exceeds that value. The neutral class is economically sensible because movements smaller than noise and trading frictions need not justify a trade.

5.2. Regularized multinomial logistic regression

The first model is logistic regression with median imputation, standardization, balanced class weights and $C = 0.2$. For class k ,

$$P(y = k \mid x) = \frac{\exp(\beta_k^\top x)}{\sum_j \exp(\beta_j^\top x)}. \quad (5.4)$$

Standardization makes coefficient magnitudes more comparable across features. The relatively strong regularization is appropriate for 101 correlated predictors and only a few thousand observations.

5.3. Histogram gradient boosting

The nonlinear model is `HistGradientBoostingClassifier`. Gradient boosting builds an additive predictor

$$F_M(x) = F_0(x) + \sum_{m=1}^M \eta h_m(x), \quad (5.5)$$

with shallow trees h_m fitted stagewise (Friedman, 2001). The configured values are 140 boosting iterations, maximum depth 3, learning rate 0.035, minimum 30 samples per leaf and L2 regularization 2.0. Class-balanced sample weights are computed from the fitting subset.

5.4. Ensemble

The final class probabilities are the arithmetic mean of the two models,

$$\hat{p}(x) = \frac{1}{2}\hat{p}_{\text{logit}}(x) + \frac{1}{2}\hat{p}_{\text{HGB}}(x). \quad (5.6)$$

No ensemble weight is tuned on validation or test data.

5.5. Why the earlier neural-network approach was not retained

A dense neural network is not intrinsically inappropriate for markets, but it adds capacity that is difficult to justify here. With roughly three thousand usable observations, 101 strongly correlated features and a noisy daily target, a high-capacity model increases variance and the opportunity for research overfitting. Logistic regression and shallow boosted trees provide nonlinear flexibility, regularization, faster repeated retraining and more transparent diagnostics. This is an engineering choice for the present sample size, not a general claim that neural networks are unsuitable for finance.

6. Experimental Methodology and Audit

6.1. Chronological train/validation/test separation

The data are never randomly shuffled. Integer boundaries are set at 60% and 80% of the usable sample: 1,887 training observations, 629 validation observations and 629 final test observations. This prevents future feature rows from entering an earlier fitting set.

6.2. Preprocessing leakage

Both median imputation and standardization for the logistic model are inside a scikit-learn Pipeline; they are therefore estimated only when `fit` is called on a particular training slice. Histogram gradient boosting uses an in-pipeline median imputer but no scaler. There is no global scaler fitted on the complete dataset. This aspect is correctly leakage-aware.

6.3. Validation-only threshold selection

The initial ensemble is fitted on the training slice and used to predict the validation slice. A predeclared 4×3 grid of confidence thresholds $\{0.45, 0.50, 0.55, 0.60\}$ and probability edges $\{0.05, 0.10, 0.15\}$ is evaluated. Candidate thresholds are considered valid only if they produce at least six position changes and at least 5% exposure. The intended score is validation Sharpe.

In the supplied run, *every candidate is invalid*; all saved scores are $-\infty$. Even the most active candidate reaches only 2.38% validation exposure. Consequently the program falls back to confidence 0.55 and edge 0.10. These are not genuinely validation-optimal thresholds. This is methodologically preferable to selecting thresholds on the test set, but it means the calibration procedure failed to identify a feasible strategy under its own minimum-activity constraints.

6.4. Walk-forward test

The test uses an expanding window and retrains every 30 observations. Before each refit at test index s , the fitting end is set to $s - 2$, so the two most recent rows are purged. This matches the two-open target maturity: at decision time, their labels would not yet be fully observed. Each model therefore predicts only future test rows relative to its fitting set.

6.5. Important residual boundary leakage

The strongest audit finding is that the *initial* model used for validation calibration is fitted on `X[:train_end]` and `y[:train_end]` without the same two-row purge. A label for row t uses O_{t+1} and O_{t+2} . Therefore the final one or two training labels at the train/validation boundary require

price information from the validation period. This creates a small but genuine label-maturity leak into the model that generates validation probabilities.

The magnitude is limited—two rows out of 1,887 training observations—and the final test walk-forward routine does purge correctly. Nevertheless, a rigorous implementation should fit the validation-calibration model on `[:train_end-2]` or more generally compute label-availability timestamps and purge programmatically. The feature-importance table is also derived from this initial model and inherits the same boundary issue.

6.6. External-market timing

Exogenous features are shifted by one day after alignment. This is a deliberate and conservative safeguard. Residual ambiguity remains because daily Yahoo timestamps abstract away exact exchange close times and because forward-filled weekend values can be several calendar days old. No obvious same-day leakage is present in the code, but exact point-in-time synchronization is not proven by the saved artifacts.

6.7. Survivorship and data-vintage considerations

The selected instruments are broad, liquid and mostly long-lived, so conventional constituent survivorship is less relevant than in a stock-universe study. However, the experiment uses current Yahoo historical series rather than a versioned point-in-time data archive. Revisions, symbol-history issues or vendor adjustments cannot be audited from the supplied outputs.

7. Trading Strategy and Backtest

7.1. Signal generation

Let the ensemble probabilities be $(p_{\downarrow}, p_0, p_{\uparrow})$. The final run is long-only. With frozen confidence threshold $\theta_c = 0.55$ and edge $\theta_e = 0.10$,

$$w_t = \begin{cases} 1, & p_{\uparrow,t} \geq \theta_c \wedge (p_{\uparrow,t} - p_{\downarrow,t}) \geq \theta_e, \\ 0, & \text{otherwise.} \end{cases} \quad (7.1)$$

The edge is measured against $p_{\downarrow}$, not against $\max(p_{\downarrow}, p_0)$. At the chosen confidence of 0.55 this distinction cannot make the neutral class exceed $p_{\uparrow}$, because the remaining two probabilities sum to at most 0.45. It would matter for lower confidence candidates in the validation grid.

7.2. Execution timing and costs

A signal based on information through close t earns the return from open $t + 1$ to open $t + 2$. The one-way friction is

$$c = \frac{7.5 + 2.5}{10,000} = 0.001, \quad (7.2)$$

that is, 10 basis points per unit of turnover. The strategy return is

$$r_t^{\text{strat}} = w_t r_t^{\text{exec}} - c |w_t - w_{t-1}|. \quad (7.3)$$

Thus an isolated one-day long position pays 10 bps on entry and 10 bps on the following transition back to cash; consecutive long days pay no additional cost between them.

7.3. Benchmarks

Two benchmarks are saved. Buy-and-hold applies the same next-open to following-open BTC return on every test observation. The trend baseline is long when both the 50-day EMA distance and 30-day BTC log return are positive. It uses the same cost engine as the ML strategy. These benchmarks are essential: a positive strategy return would not by itself demonstrate value, and a lower loss in a bear phase may be explained by reduced exposure rather than superior forecasting.

7.4. Performance metrics

For daily strategy return r_t and equity $V_T = \prod_t(1 + r_t)$, the implementation uses

$$R_{\text{tot}} = V_T - 1, \quad (7.4)$$

$$\text{CAGR} = V_T^{365/n} - 1, \quad (7.5)$$

$$\sigma_{\text{ann}} = \text{std}(r_t) \sqrt{365}, \quad (7.6)$$

$$S = \frac{\text{mean}(r_t)}{\text{std}(r_t)} \sqrt{365}, \quad (7.7)$$

$$DD_t = \frac{V_t}{\max_{\tau \leq t} V_\tau} - 1, \quad (7.8)$$

$$\text{MDD} = \min_t DD_t, \quad (7.9)$$

$$\text{Calmar} = \frac{\text{CAGR}}{|\text{MDD}|}. \quad (7.10)$$

The Sharpe ratio uses a zero risk-free rate and population standard deviation (`ddof=0`). Exposure is the fraction of days with nonzero position, and `position_changes` counts days on which the position changes.

8. Results

8.1. Classification performance

The strict test contains 629 observations: 228 down, 167 neutral and 234 up. The argmax predictions contain 252 down, 173 neutral and 204 up. Balanced accuracy is 0.3772. For three classes, a reference balanced accuracy under class-symmetric guessing is approximately $1/3 = 0.3333$, so the gain is only 4.39 percentage points. Test log loss is 1.0955, barely lower than the uniform-probability reference $-\log(1/3) = 1.0986$. The classifier therefore shows weak, not strong, multiclass predictive skill.

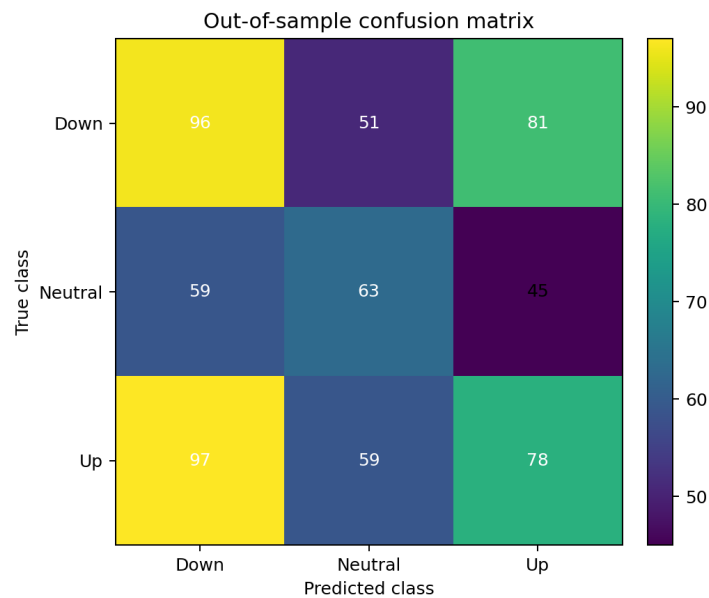


Figure 8.1.: Out-of-sample confusion matrix. No class is cleanly separated; recall is 42.1% for down, 37.7% for neutral and 33.3% for up.

8.2. Economic performance

Table 8.1.: Strict out-of-sample performance. Percentages and ratios are taken from the supplied metrics.

Model	Total	CAGR	Vol.	Sharpe	Max DD	Calmar	Exposure	Changes
ML strategy	-3.75%	-2.19%	15.21%	-0.068	-20.67%	-0.106	1.59%	14
Buy & hold	-19.24%	-11.66%	43.67%	-0.065	-53.06%	-0.220	—	—
Trend baseline	-7.97%	-4.70%	22.03%	-0.109	-26.50%	-0.177	36.57%	43

The ML strategy loses less capital than both benchmarks, but its Sharpe ratio is still negative and almost identical to buy-and-hold. The crucial number is 1.59% exposure: only 10 of 629 test days have a nonzero ML position. This makes the result primarily one of avoiding market participation. Outperforming buy-and-hold during a falling test period is therefore not, by itself, evidence of alpha.

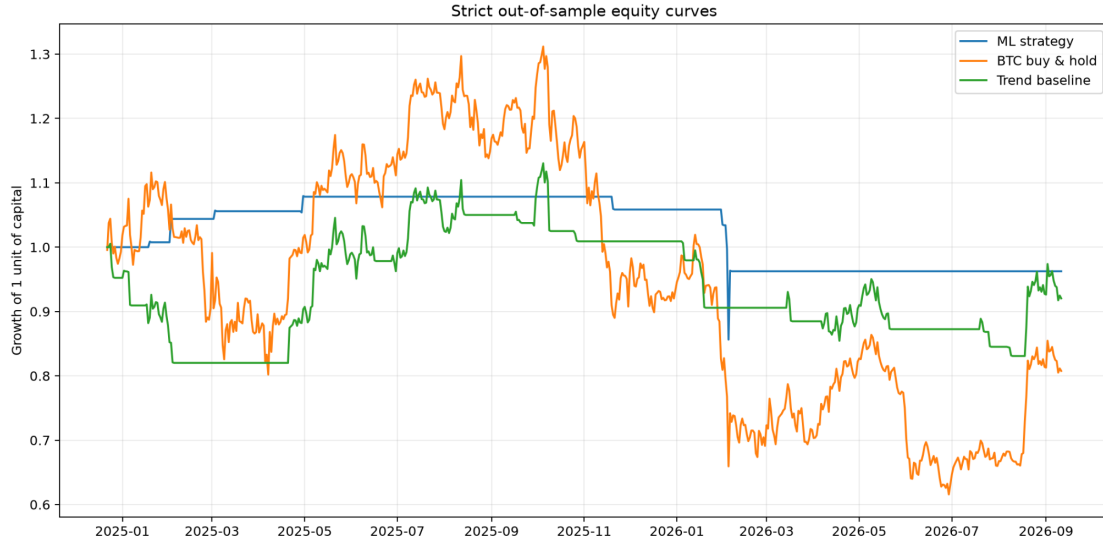


Figure 8.2.: Strict out-of-sample equity curves. The ML strategy preserves more capital than buy-and-hold and the trend baseline, but this must be interpreted together with only 1.59% exposure; the curve is flat mainly because the strategy is usually in cash.

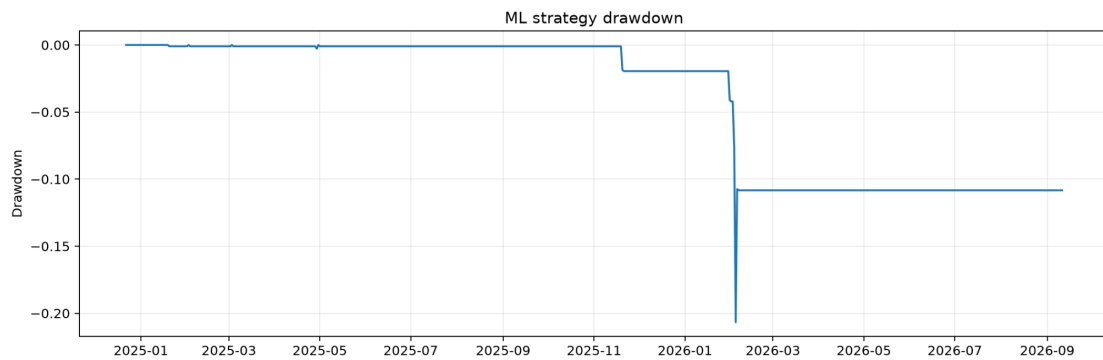


Figure 8.3.: ML-strategy drawdown. Most of the test period is flat, followed by a concentrated loss episode around early February 2026; maximum drawdown reaches 20.67%.

8.3. Prediction confidence

The median of the maximum class probability is 0.4243. The 90th, 95th and 99th percentiles are 0.5180, 0.5494 and 0.6231. Thus fewer than roughly 5% of observations even reach the final confidence threshold of 0.55 before the additional up-versus-down edge condition is applied. Strong-confidence predictions are rare.

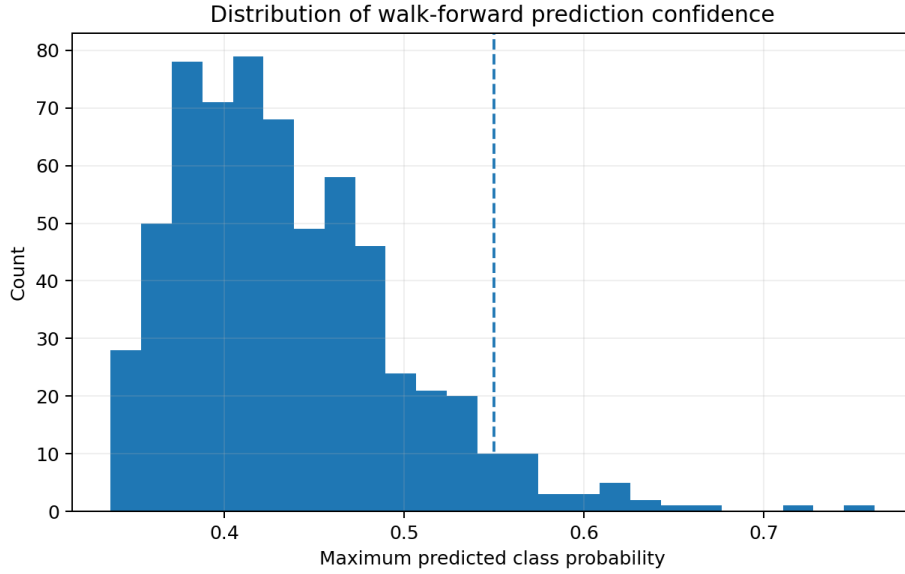


Figure 8.4.: Distribution of maximum class probability. The dashed line marks the 0.55 trading-confidence threshold. Most walk-forward predictions remain close to the three-class uncertainty region rather than becoming highly concentrated.

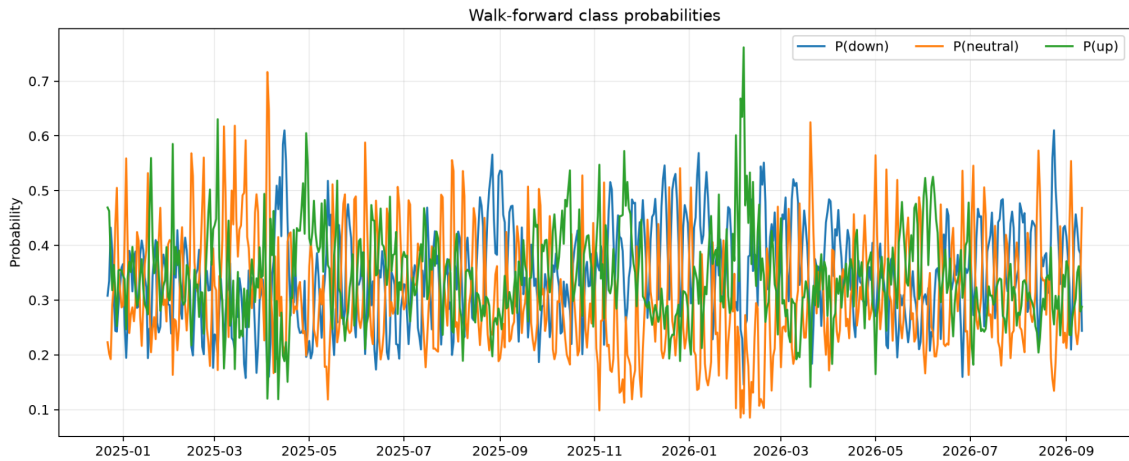


Figure 8.5.: Walk-forward class probabilities. Most probabilities fluctuate near one third, with intermittent confidence spikes. This pattern is consistent with weak average directional separation and a small number of high-confidence trading signals.

8.4. Feature importance

The saved logistic-regression importance is the mean absolute coefficient across classes. Because the logistic pipeline standardizes all features before fitting, coefficient magnitudes are more comparable than they would be on raw scales. The five largest values are BTC 60-day log return (0.2030), BTC RSI(14) (0.2029), BTC 20-day drawdown (0.1881), BTC 30-day realized volatility (0.1849), and SPY 60-day log return (0.1797). The leading variables span momentum, oscillator, drawdown, volatility and cross-asset equity state rather than a single economic mechanism.

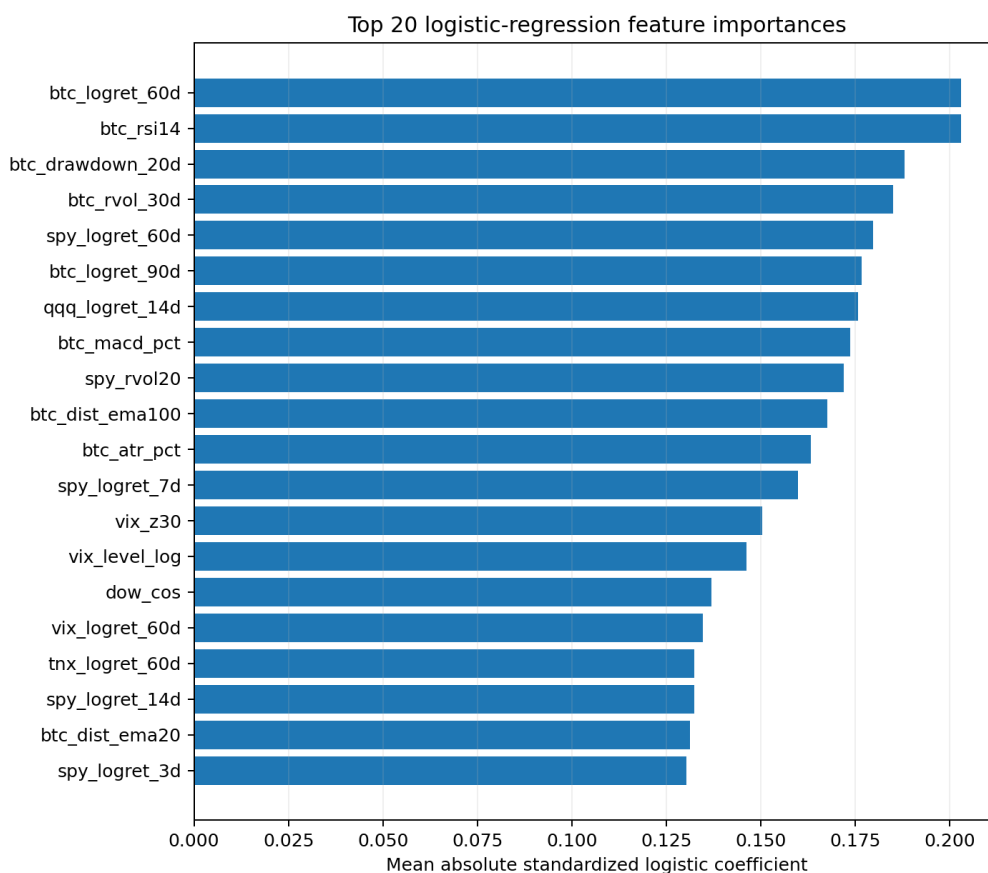


Figure 8.6.: Top 20 mean absolute standardized logistic coefficients from the initial training model. Importance is descriptive, not proof of stable causal or predictive value; coefficient rankings can change across walk-forward refits.

9. Interpretation

9.1. Directional prediction versus regime detection

The additional diagnostics reveal a clear asymmetry. One-vs-rest AUC is 0.5233 for down, 0.5267 for up and 0.6014 for neutral. Equivalently, using $1 - p_0$ to rank significant versus neutral movements gives AUC 0.6014. By contrast, if neutral observations are removed and direction is evaluated only on significant moves using $p_{\uparrow}/(p_{\uparrow} + p_{\downarrow})$, the AUC is 0.4838, below the no-skill level of 0.5.

This supports the hypothesis that the model is better at distinguishing quiet versus significant movement than at predicting the sign of that movement. The result is not merely visual. p_0 has Pearson correlation -0.1910 and Spearman correlation -0.1986 with the absolute next open-to-open return: higher predicted neutral probability is associated with smaller realized movement amplitude.

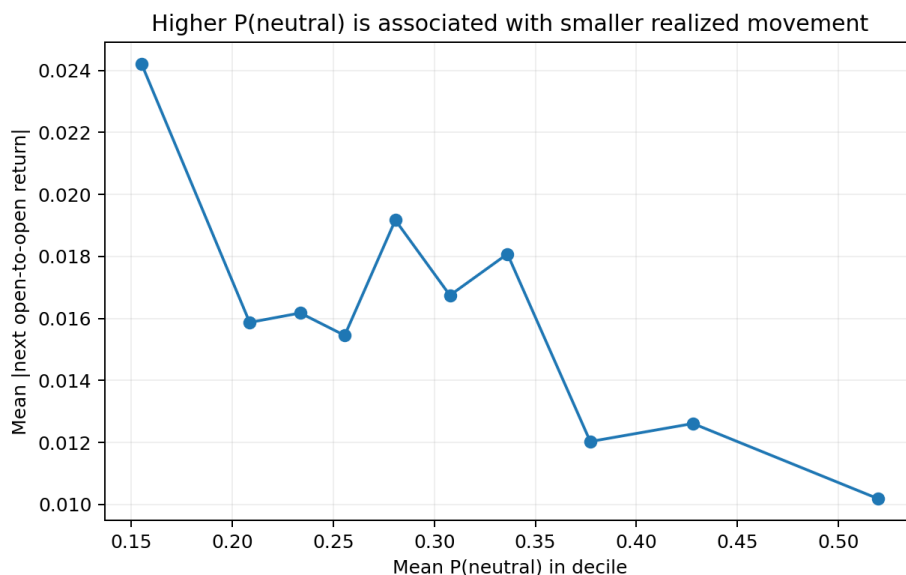


Figure 9.1.: Mean absolute next-period BTC return across deciles of predicted neutral probability. The relationship is noisy but generally negative, consistent with the neutral classifier carrying more regime/amplitude information than the directional probabilities.

9.2. Forensic analysis of the maximum drawdown

The strategy's equity peak is 1.0795 on 30 April 2025. Maximum drawdown occurs on 4 February 2026, when equity reaches 0.8563, corresponding to -20.67% from the prior peak. The immediate cause is not a long sequence of small errors but a concentrated cluster of high-confidence long signals during a violent decline.

Table 9.1.: Active test days. Returns include entry/exit friction as implemented; on consecutive long days, no additional turnover cost is charged between days.

Date	$p_{\downarrow}$	p_0	$p_{\uparrow}$	True	BTC ret.	Strat. ret.	Equity
2025-01-19	0.236	0.205	0.560	Up	0.958%	0.858%	1.009
2025-02-02	0.251	0.163	0.585	Up	3.806%	3.706%	1.045
2025-03-03	0.197	0.172	0.630	Up	1.347%	1.247%	1.057
2025-04-29	0.196	0.199	0.605	Neutral	-0.078%	-0.178%	1.054
2025-04-30	0.205	0.244	0.551	Up	2.422%	2.422%	1.079
2025-11-20	0.315	0.113	0.572	Down	-1.653%	-1.753%	1.059
2026-01-31	0.296	0.102	0.601	Down	-2.108%	-2.208%	1.035
2026-02-03	0.246	0.085	0.668	Down	-3.469%	-3.569%	0.997
2026-02-04	0.230	0.136	0.635	Down	-14.123%	-14.123%	0.856
2026-02-05	0.146	0.093	0.762	Up	12.518%	12.518%	0.964

The 4 February observation is especially revealing: $p_{\uparrow} = 0.635$ while $p_{\downarrow} = 0.230$, yet the realized open-to-open return is -14.12%. The following day the model becomes even more bullish ($p_{\uparrow} = 0.762$) and the market rebounds +12.52%. The model therefore appears to recognize that the regime is unusually non-neutral— p_0 is very low on both days—while failing to resolve direction reliably. This is consistent with the aggregate AUC results.

9.3. Does the strategy demonstrate alpha?

No. Nominally, the ML strategy loses less than the benchmarks over this particular test sample. Risk-adjusted performance is still negative, and after accounting for the 1.59% exposure the benchmark comparison is not evidence that the classifier generates positive excess return per unit of risk or capital deployed. No formal data-snooping-adjusted hypothesis test, bootstrap reality check or independent second holdout is supplied. The strongest defensible conclusion is that there is weak regime information and insufficient evidence of a robust directional trading edge.

10. Limitations

1. **Train/validation label-maturity leakage.** The initial calibration model is not purged by two rows at the train/validation boundary even though its labels require two future opens. This is a real implementation defect, albeit affecting only two of 1,887 training rows.
2. **Failed threshold calibration.** All 12 validation candidates violate the minimum exposure/trade-count criteria and receive $-\infty$ scores. Final thresholds 0.55/0.10 are fallback defaults, not validation-optimal parameters.
3. **Small effective number of trades.** The ML strategy is active on only 10 test days and experiences 14 position changes. Financial conclusions from such a sparse strategy are statistically fragile.
4. **Limited number of independent regimes.** There are 629 daily test observations but far fewer independent macro/crypto regimes. Serial dependence further reduces effective sample size.
5. **Non-stationarity.** Relationships between crypto, equities, volatility, rates and the dollar can change structurally.
6. **Feature dimensionality and collinearity.** 101 features include many overlapping horizons and related indicators. Regularization reduces but does not eliminate selection variance.
7. **Probability calibration.** No dedicated calibration method such as isotonic regression or Platt-style post-calibration is used, and calibration quality is not separately validated.
8. **Hyperparameter provenance.** The current run keeps model hyperparameters fixed during validation/test, but the artifacts cannot prove that earlier human iteration never used the same final test period. Repeated research on the same holdout can turn it into a de-facto validation set (White, 2000; Bailey et al., 2017).
9. **Daily execution model.** The backtest assumes fills at next daily open with fixed 7.5 bps fee and 2.5 bps slippage. Spread, market impact, exchange outages and order-size dependence are ignored.
10. **Long/cash only.** The saved experiment does not test shorting, leverage or risk-targeted position sizes.
11. **Market-hours synchronization.** A one-day lag is conservative, but exact timestamp semantics across BTC and traditional markets are not established from a point-in-time feed.
12. **Data-vintage risk.** Yahoo historical data are downloaded retrospectively. The experiment does not preserve an immutable contemporaneous data snapshot.
13. **Missing information sets.** No order-book, funding-rate, futures-basis, options-implied-volatility or on-chain variables are present.
14. **Single primary asset.** Results are for BTC only and do not establish transferability.
15. **No independent second final holdout.** Because this report itself analyzes the test period in detail, future model revisions should not continue to evaluate on the same

2024–2026 interval as if it were untouched.

11. Future Work

11.1. Fix the evaluation protocol before increasing model complexity

The first revision should be methodological, not architectural. Purge label-immature rows at *every* fitting boundary, including the initial train/validation fit. Encode label-availability times explicitly instead of relying on a hard-coded two-row rule. Replace the current threshold grid or its feasibility constraints so that validation can actually select among valid candidates. Then reserve a new, genuinely untouched final holdout.

11.2. Hierarchical regime–direction–risk framework

The observed AUC pattern suggests a two- or three-stage model that matches the information actually present in the data.

Stage 1: movement/regime model.

$$q_t = P(|r_{t+1}| > b_t \mid x_t). \quad (11.1)$$

The present experiment already suggests nontrivial information in this task because $1 - p_0$ achieves AUC 0.601.

Stage 2: conditional direction. Only when the first stage predicts a significant move should a second classifier estimate

$$d_t = P(r_{t+1} > 0 \mid |r_{t+1}| > b_t, x_t). \quad (11.2)$$

The current conditional directional AUC of 0.484 shows that this stage requires new information or a different representation; merely increasing capacity is not justified by the present evidence.

Stage 3: risk sizing. A confidence and volatility-aware position can be defined as

$$w_t = \text{clip} \left(s_t c_t \frac{\sigma_{\text{target}}}{\hat{\sigma}_t}, -w_{\max}, w_{\max} \right), \quad (11.3)$$

where s_t is the directional sign, c_t a calibrated confidence score and $\hat{\sigma}_t$ estimated forward risk. Such sizing should be validated separately and must not be optimized on the final holdout.

11.3. Other justified extensions

Promising experiments include nested walk-forward validation, rolling versus expanding windows, probability calibration, feature-stability analysis across refits, aggressive removal of redundant horizons, multiple forecast horizons, regime-specific models, bootstrap uncertainty intervals, and new information sets such as futures basis, perpetual funding, options implied volatility and on-chain activity. None is guaranteed to improve performance; each should be evaluated under the same leakage-aware protocol.

12. Conclusion

This experiment is methodologically stronger than a conventional random-split financial ML backtest. Features are causal, preprocessing is fitted inside model pipelines, exogenous daily series are conservatively lagged, strategy thresholds are not tuned on the final test set, execution is delayed to the next open, trading frictions are included, and the final test is generated through expanding-window walk-forward refits.

The numerical result nevertheless does not support a robust directional trading edge. Test balanced accuracy is 0.3772 and log loss 1.0955, both only marginally better than simple three-class reference values. The long/cash ML strategy loses 3.75%, versus -19.24% for buy-and-hold and -7.97% for the trend baseline, but it is exposed on only 1.59% of test days. Its lower cumulative loss is therefore mostly risk avoidance, not demonstrated alpha.

The more informative finding is structural. Neutral-class AUC is 0.601, while up and down AUCs are close to 0.52, and conditional directional AUC on significant moves is 0.484. Predicted neutral probability is negatively associated with realized absolute return. The large-loss episode reinforces the same interpretation: the model becomes highly confident that the market is non-neutral but can choose the wrong sign during violent moves.

Two methodological corrections are essential before further claims: purge the two label-immature observations at the initial train/validation boundary, and redesign the validation threshold procedure because every grid candidate currently fails its own feasibility constraints, forcing fallback thresholds. After these corrections, the most defensible next step is not a larger directional classifier but a hierarchical regime–direction–risk framework tested on a new untouched holdout.

A. Complete Configuration

```
primary: BTC-USD
start: 2018-01-01
end: null
exogenous: [ETH-USD, SPY, QQQ, GLD, ^VIX, DX-Y.NYB, ^TNX]
train_fraction: 0.60
validation_fraction: 0.20
neutral_vol_multiplier: 0.25
min_neutral_return: 0.004
fee_bps: 7.5
slippage_bps: 2.5
retrain_every: 30
long_only: true
random_state: 42
hgb_max_iter: 140
hgb_max_depth: 3
hgb_learning_rate: 0.035
hgb_min_samples_leaf: 30
hgb_l2: 2.0
logistic_c: 0.2
```

B. Complete Feature List

The following list is generated from the supplied logistic-importance file and therefore corresponds exactly to the 101 fitted feature columns in the saved run.

No.	Feature	Importance
1	btc_logret_60d	0.2030
2	btc_rsi14	0.2029
3	btc_drawdown_20d	0.1881
4	btc_rvol_30d	0.1849
5	spy_logret_60d	0.1797
6	btc_logret_90d	0.1767
7	qqq_logret_14d	0.1757
8	btc_macd_pct	0.1736
9	spy_rvol20	0.1718
10	btc_dist_ema100	0.1676
11	btc_atr_pct	0.1631
12	spy_logret_7d	0.1598
13	vix_z30	0.1502
14	vix_level_log	0.1461
15	dow_cos	0.1370
16	vix_logret_60d	0.1345
17	tnx_logret_60d	0.1324
18	spy_logret_14d	0.1324
19	btc_dist_ema20	0.1311
20	spy_logret_3d	0.1303
21	gld_logret_7d	0.1290
22	btc_dist_ema50	0.1250
23	btc_rvol_14d	0.1249
24	tnx_z30	0.1214
25	dx_y_nyb_logret_14d	0.1186
26	btc_logret_7d	0.1121
27	tnx_logret_1d	0.1075
28	tnx_logret_14d	0.1047
29	qqq_logret_3d	0.1028
30	tnx_level_log	0.1027
31	btc_dist_ema200	0.0994
32	dx_y_nyb_z30	0.0990
33	qqq_logret_60d	0.0954
34	eth_usd_level_log	0.0947
35	dx_y_nyb_logret_60d	0.0924
36	vix_logret_1d	0.0922
37	eth_usd_logret_60d	0.0889
38	qqq_z30	0.0884
39	btc_bb_z20	0.0871

No.	Feature	Importance
40	tnx_rvol20	0.0865
41	qqq_logret_7d	0.0861
42	gld_z30	0.0854
43	btc_macd_hist_pct	0.0819
44	gld_level_log	0.0815
45	btc_minus_spy_mom30	0.0811
46	btc_drawdown_180d	0.0804
47	btc_volume_change_7d	0.0800
48	dow_sin	0.0799
49	qqq_level_log	0.0797
50	vix_rvol20	0.0779
51	btc_dist_ema10	0.0758
52	dx_y_nyb_logret_30d	0.0753
53	gld_logret_30d	0.0746
54	eth_usd_logret_14d	0.0742
55	eth_usd_rvol20	0.0734
56	btc_logret_2d	0.0719
57	gld_logret_60d	0.0704
58	btc_bb_width20	0.0697
59	gld_logret_3d	0.0694
60	btc_drawdown_60d	0.0670
61	spy_logret_30d	0.0659
62	spy_logret_1d	0.0659
63	btc_rvol_7d	0.0658
64	eth_usd_logret_7d	0.0637
65	eth_usd_z30	0.0633
66	vix_logret_7d	0.0622
67	gld_logret_14d	0.0619
68	dx_y_nyb_logret_3d	0.0584
69	btc_logret_3d	0.0579
70	qqq_logret_1d	0.0567
71	btc_logret_1d	0.0566
72	spy_level_log	0.0565
73	btc_logret_14d	0.0539
74	btc_rvol_60d	0.0511
75	tnx_logret_7d	0.0485
76	btc_volume_change_1d	0.0479
77	qqq_logret_30d	0.0461
78	eth_usd_logret_3d	0.0441
79	dx_y_nyb_logret_7d	0.0422
80	btc_range_pct	0.0418
81	month_sin	0.0414
82	tnx_logret_3d	0.0388
83	btc_minus_eth_mom30	0.0380
84	qqq_rvol20	0.0365
85	dx_y_nyb_level_log	0.0358
86	eth_usd_logret_1d	0.0350
87	vix_logret_14d	0.0343
88	month_cos	0.0297
89	btc_volume_z30	0.0274

No.	Feature	Importance
90	spy_z30	0.0273
91	eth_usd_logret_30d	0.0270
92	btc_intraday_ret	0.0270
93	gld_logret_1d	0.0244
94	gld_rvol20	0.0239
95	btc_logret_30d	0.0238
96	vix_logret_3d	0.0228
97	dx_y_nyb_rvol20	0.0220
98	btc_gap	0.0203
99	dx_y_nyb_logret_1d	0.0200
100	vix_logret_30d	0.0148
101	tnx_logret_30d	0.0137

C. Validation Threshold Grid

Table C.1.: Validation grid. Every candidate has score $-\infty$ because no candidate simultaneously satisfies at least six position changes and at least 5% exposure.

Confidence	Edge	Exposure	Changes	Total return	Sharpe
0.60	0.05	0.0000	0	0.0000	–
0.60	0.10	0.0000	0	0.0000	–
0.60	0.15	0.0000	0	0.0000	–
0.55	0.05	0.0016	2	0.0051	0.629
0.55	0.10	0.0016	2	0.0051	0.629
0.55	0.15	0.0016	2	0.0051	0.629
0.50	0.05	0.0079	10	-0.0196	-0.134
0.50	0.10	0.0079	10	-0.0196	-0.134
0.50	0.15	0.0079	10	-0.0196	-0.134
0.45	0.10	0.0223	12	-0.0674	-0.419
0.45	0.15	0.0207	12	-0.0982	-0.677
0.45	0.05	0.0238	14	-0.0794	-0.502

D. Reproducibility and Execution

The original source file is preserved in `data/btc_ml_research.py`; supplied CSV/JSON outputs are preserved in `data/`. The experiment requires Python, NumPy, pandas, matplotlib, scikit-learn and yfinance. A typical environment is:

```
python -m venv .venv
# Windows PowerShell:
.\.venv\Scripts\Activate.ps1
pip install numpy pandas matplotlib scikit-learn yfinance
python btc_ml_research.py --output-dir outputs
```

For a strict future replication, archive the downloaded raw data as well as the code and outputs, because a later Yahoo download may not be byte-identical to the historical data used in this run.

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